Theory of Automata & Formal Languages
Assignment-02 (Spring 2026)

Submission should be in Handwritten Form
Due Date: Friday, April 03, 2026

INSTRUCTIONS:

1. Solve each problem on a separate sheet of paper (using pen/ball pen). Write your Name, Registration Number and signature on every sheet used.
2. Take a picture of your sheet(s) and paste it under the question statement in the word document.
3. Submit the handwritten PDF document on the portal.

CLO	CLO STATEMENT	Bloom's Taxonomy Level	PLO
1	Understand basics of formal language classes and prove language membership properties with focus regular languages using State machines.	C2	2

Task 1: Regular Expression and Construction of DFA

Construct a DFA for the following languages.

a) $L = \{w \in \Sigma^* \mid w \text{ does not contain an equal number of occurrences of the substrings } 01 \text{ and } 10\}$
with $\Sigma = \{0, 1\}$

b) $L = \{w \in \{0, 1\}^* \mid w \text{ contains exactly one occurrence of the substring } 01\}$

c) $L = \{w \mid w \text{ every odd position of } w \text{ is } a\}$, $w \in \{a, b\}^*$

d) John C Martin Exercise 2.1, 2.12; Sipser Exercise 1.4, 1.5, 1.6, 1.7

e) $L = \{s \text{ over } \Sigma = \{0,1,2\} \mid \text{number in } s \text{ is divisible by } 5 \text{ and } 10\}$

f) $L = \{s \text{ over } \Sigma = \{0,1,2\} \mid \text{sum of digits in } s \text{ is divisible by } 5 \text{ and } 10\}$

g) $L = \{s \text{ over } \Sigma = \{0,1\} \mid \text{binary string in } s \text{ is divisible by } 5 \text{ and } 10\}$

h) $L = \{w \in \{a, b\}^* \mid w \text{ ends with } abb\}$

i) $L = \{w \in \{a, b\}^* \mid w \text{ contain substring } baabba\}$

j) $L = \{w \in \{a, b\}^* \mid w \text{ contains atleast three } b's\}$

k) $L = \{w \in \{0,1,2,3\}^* \mid \text{sum of digits in } w \text{ are either divisible by } 2 \text{ or } 4\}$

l) $L = \{w \in \{a, b, c\}^* \mid n_a(w) \text{ and } n_b(w) \text{ are even but } n_c(w) \text{ are odd}\}$

m) $L = \{w \in \{0,1\}^* \mid w \text{ contains the substring } 010, \text{but does not contain the substring } 0101\}$

n) $L = \{w \in \{a, b\}^* \mid w \text{ start with } 'a' \&\& \text{ contain even number of } b's \text{ OR end with } 'b' \&\& \text{ contain odd number of } a$

o) $L = \{w \in \{a, b\}^* \mid w \text{ start with } abb\}$

p) $L = \{w \in \{a, b\}^* \mid w \text{ contain substring } abbaab\}$

q) $L = \{w \in \{a, b\}^* \mid w \text{ contain at most one time substring } abbaab\}$

r) $L = \{w \in \{a, b\}^* \mid w \text{ contains substring three } a's\}$

s) $L = \{w \in \{a, b\}^* \mid w \text{ contains at most three } a's\}$

t) $L = \{w \in \{a, b\}^* \mid w \text{ contains exact three } a's\}$

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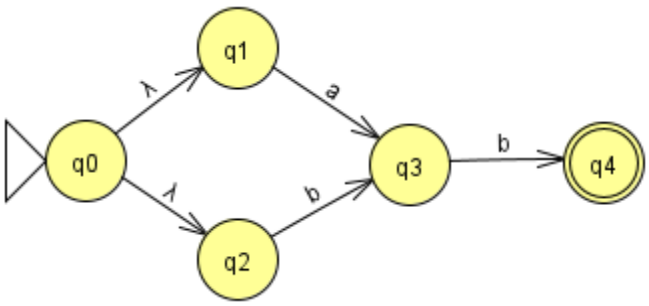
u) $L = \{w \in \{a,b\}^* | In\ w\ every\ a\ (if\ there\ are\ any)\ is\ followed\ immediately\ by\ bb\}$
v) $L = \{w \in \{a,b\}^* | w\ \ containing\ both\ bb\ and\ aba\ as\ substrings\ \}$
w) $L = \{w \in \{a,b\}^* | w\ \ containing\ bb\ or\ aba\ as\ substrings\ \}$
x) $L = \{w \in \{a,b\}^* | w\ \ containing\ both\ aba\ and\ bab\ as\ substrings\ \}$
y) $L = \{w \in \{0,1\}^* | every\ 0\ in\ w\ is\ followed\ by\ at\ least\ one\ 1\}$
z) $\boldsymbol{L} = \{w \in \{a,b\}^* | w\ contains\ substring\ abb\}$
aa) $\boldsymbol{L} = \{w \in \{a,b\}^* | w\ contains\ atleast\ three\ a's\}$
bb) $\boldsymbol{L} = \{w \in \{0,1\}^* | every\ 0\ in\ w\ is\ followed\ by\ at\ least\ one\ 1\}$
cc) $\boldsymbol{L} = \{w \mid w\ over\ \Sigma = \{0,1\}, w\ contains\ 10\ but\ does\ not\ end\ with\ 101\}$
dd) $\boldsymbol{L} = \{w \in \{a,b,c\}^* | a^i b^j c^k where\ i\ ,j,k \geq 0\}$
ee) $\boldsymbol{L} = \{w \in \{a,b,c\}^* | a^i b^j c^k where\ i\ ,j,k > 0\}$
ff) $\boldsymbol{L} = \{aba\{aa + b\}^*\{ba\}^* b\}$
gg) $\boldsymbol{L} = \{aba\{ba\}^* b\}$
hh) $L_1 = \{w \in \{0,1\}^* \mid w\ ends\ with\ 00\}\ with\ three\ states$
ii) $L_2 = \{w \in \{0,1\}^* \mid w\ contains\ substring\ 0101\ i.e. w = x0101y\}\ with\ five\ states\ states$
jj) $L_3 = \{w \in \{0,1\}^* \mid w\ ends\ with\ 00\ or\ ends\ 11\}\ with\ five\ states$
kk) $L_4 = \{w \in \{0,1\}^* \mid w\ contains\ substring\ 00\ or\ \ substring\ 11\}\ with\ five\ states\ states$

Task 2 : Construction of NFA – Conversion of NFA - DFA

a) Consider the regular expression (01 + 010) *. Give a 3-state NFA that recognizes the same language. Give a DFA that recognizes the same language by converting your constructed NFA to a DFA. Σ = {0 , 1}

b) Convert the NFA with λ into its equivalent DFA. Σ = {a ,b}



c) Convert the NFA with λ into its equivalent DFA. Σ = {a ,b, c}

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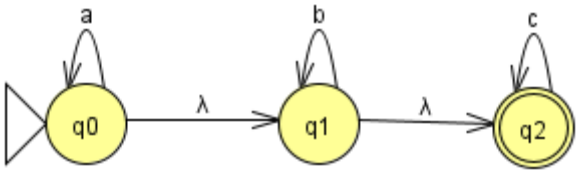

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